



Unit II: Random Variables

Discrete & Continuous Random Variables,
Probability Mass/Density Functions,
Cumulative Distribution Functions,
Expectation and Variance

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1 Motivation: From Events to Random Variables

In Unit I, we studied probability by defining sample spaces, events, and probability measures. For example, when tossing two coins, the sample space was

$$S = \{HH, HT, TH, TT\},$$

and we could compute the probability of events like “at least one head” by counting favourable outcomes.

However, in engineering practice, we are often more interested in **numbers** associated with the outcomes rather than the outcomes themselves. Consider the following questions:

- How many defective items are there in a batch of 100?
- What is the voltage across a resistor at any instant?
- How long does a machine operate before failure?
- What is the error rate in a communication channel?

In each case, the answer is a real number. We can describe the experiment perfectly well by the sample space, but our interest is in a numerical summary. This leads to the concept of a **random variable**.

Key idea: A random variable is a function that assigns a real number to each outcome of a random experiment. It converts the abstract sample space into the real line, where we can apply all the tools of calculus and algebra.

For example, in the two-coin toss:

$$X = \text{number of heads} = \begin{cases} 0 & \text{if } TT \\ 1 & \text{if } HT \text{ or } TH \\ 2 & \text{if } HH \end{cases}$$

Now instead of asking $P(\{HH\})$, we ask $P(X = 2)$. Both mean the same thing. The benefit is that we can now talk about $P(X \leq 2)$, $P(0 \leq X \leq 1)$, and compute averages and variances.

2 Definition of a Random Variable

A **random variable** X is a real-valued function defined on the sample space S . Formally,

$$X : S \rightarrow \mathbb{R}.$$

For each outcome $s \in S$, the random variable assigns a number $X(s)$.

Notation:

- Capital letters X, Y, Z denote random variables.
- Lowercase letters x, y, z denote the values they take.
- $P(X = x)$ means “the probability that the random variable X takes the value x ”.

Random variables are of two fundamental types: **discrete** and **continuous**.

3 Discrete Random Variables

A random variable is **discrete** if its set of possible values is finite or countably infinite. In practice, this means it takes values that can be listed: $x_1, x_2, x_3, \dots$ Examples:

- Number of defective items in a sample.
- Number of bits in error in a data packet.
- Number of customers arriving in a queue per hour.

3.1 Probability Mass Function (PMF)

For a **discrete** random variable X , the **probability mass function** $p(x)$ is defined as

$$p(x) = P(X = x).$$

It satisfies:

1. $p(x) \geq 0$ for all x .
2. $\sum_x p(x) = 1$ (sum over all possible values).

Solved Example PMF of two-coin toss

Let X = number of heads in two tosses of a fair coin. Find the PMF.

Solution:

$$P(X = 0) = P(TT) = \frac{1}{4}, \quad P(X = 1) = P(HT \text{ or } TH) = \frac{2}{4} = \frac{1}{2},$$

$$P(X = 2) = P(HH) = \frac{1}{4}.$$

Thus the PMF is:

$$p(0) = \frac{1}{4}, \quad p(1) = \frac{1}{2}, \quad p(2) = \frac{1}{4}.$$

Check: $\sum p(x) = \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1$.

Solved Example PMF of defective items

A box contains 20 fuses, of which 5 are defective. Three fuses are drawn without replacement. Let X be the number of defective fuses drawn. Find and sketch the PMF.

Solution: X can take values 0, 1, 2, 3. This is a hypergeometric experiment (discussed later):

$$p(k) = \frac{\binom{5}{k} \binom{15}{3-k}}{\binom{20}{3}}, \quad k = 0, 1, 2, 3.$$

Calculations:

$$\binom{20}{3} = 1140.$$

$$p(0) = \frac{\binom{15}{3}}{1140} = \frac{455}{1140} \approx 0.3991.$$

$$p(1) = \frac{\binom{5}{1} \binom{15}{2}}{1140} = \frac{5 \times 105}{1140} = \frac{525}{1140} \approx 0.4605.$$

$$p(2) = \frac{\binom{5}{2} \binom{15}{1}}{1140} = \frac{10 \times 15}{1140} = \frac{150}{1140} \approx 0.1316.$$

$$p(3) = \frac{\binom{5}{3}}{1140} = \frac{10}{1140} \approx 0.0088.$$

Check sum: $0.3991 + 0.4605 + 0.1316 + 0.0088 = 1.0000$.

Solved Example PMF verification

Verify whether the following can serve as a PMF:

$$p(x) = \frac{x}{10}, \quad x = 1, 2, 3, 4.$$

Solution:

$$p(1) = 0.1, \quad p(2) = 0.2, \quad p(3) = 0.3, \quad p(4) = 0.4.$$

Sum = $0.1 + 0.2 + 0.3 + 0.4 = 1.0$, and each $p(x) \geq 0$. Hence it is a valid PMF.

Solved Example PMF of sum of two dice

Two fair dice are rolled. Let X be the sum of the two numbers. Find the PMF of X .

Solution: The possible values are $2, 3, \dots, 12$. Counting outcomes:

Sum x	Number of outcomes	$p(x)$
2	1	$1/36$
3	2	$2/36 = 1/18$
4	3	$3/36 = 1/12$
5	4	$4/36 = 1/9$
6	5	$5/36$
7	6	$6/36 = 1/6$
8	5	$5/36$
9	4	$1/9$
10	3	$1/12$
11	2	$1/18$
12	1	$1/36$

Sum of probabilities = 1. This PMF is symmetric about 7.

Solved Example Finding a constant in a PMF

Let X be a discrete random variable with PMF $p(x) = c(x+1)$ for $x = 0, 1, 2, 3$, and zero otherwise. Find the constant c .

Solution: We require $\sum_{x=0}^3 p(x) = 1$.

$$p(0) = c(1), \quad p(1) = c(2), \quad p(2) = c(3), \quad p(3) = c(4).$$

$$c(1 + 2 + 3 + 4) = 10c = 1 \implies c = \frac{1}{10}.$$

4 Continuous Random Variables

A random variable is **continuous** if it can take any value in an interval or union of intervals. Examples:

- The lifetime of a component (in hours).
- The voltage across a resistor.
- The height of a student (measured to arbitrary precision).

4.1 Probability Density Function (PDF)

For a **continuous** random variable X , the **probability density function** $f(x)$ is a function such that

$$P(a \leq X \leq b) = \int_a^b f(x) dx.$$

It satisfies:

1. $f(x) \geq 0$ for all x .
2. $\int_{-\infty}^{\infty} f(x) dx = 1$.

Important: For a continuous random variable, $P(X = a) = 0$ for any single point a . Probabilities are defined over intervals only.

Solved Example PDF verification

Consider the function $f(x) = 2x$, $0 \leq x \leq 1$ (and zero elsewhere). Verify whether it is a valid PDF.

Solution:

$$\int_{-\infty}^{\infty} f(x) dx = \int_0^1 2x dx = [x^2]_0^1 = 1.$$

Also $f(x) = 2x \geq 0$ on $[0, 1]$. Hence it is a valid PDF.

Solved Example Probability from PDF

For the PDF $f(x) = 2x$, $0 \leq x \leq 1$, find $P(0.2 \leq X \leq 0.6)$.

Solution:

$$P(0.2 \leq X \leq 0.6) = \int_{0.2}^{0.6} 2x \, dx = [x^2]_{0.2}^{0.6} = 0.36 - 0.04 = 0.32.$$

Solved Example Find k for a valid PDF

Find the value of k such that $f(x) = kx^2$, $0 \leq x \leq 2$, is a valid PDF.

Solution:

$$\int_0^2 kx^2 \, dx = k \left[\frac{x^3}{3} \right]_0^2 = k \cdot \frac{8}{3} = 1 \Rightarrow k = \frac{3}{8}.$$

Solved Example Uniform PDF

A random variable X has PDF

$$f(x) = \begin{cases} \frac{1}{b-a}, & a \leq x \leq b, \\ 0, & \text{otherwise.} \end{cases}$$

Show that this is a valid PDF and find $P\left(\frac{a+b}{2} \leq X \leq b\right)$.

Solution:

$$\int_{-\infty}^{\infty} f(x) \, dx = \int_a^b \frac{1}{b-a} \, dx = \frac{b-a}{b-a} = 1.$$

For the probability:

$$P\left(\frac{a+b}{2} \leq X \leq b\right) = \int_{\frac{a+b}{2}}^b \frac{1}{b-a} \, dx = \frac{b - \frac{a+b}{2}}{b-a} = \frac{\frac{b-a}{2}}{b-a} = \frac{1}{2}.$$

Solved Example Triangular PDF

Consider the triangular PDF

$$f(x) = \begin{cases} x, & 0 \leq x \leq 1, \\ 2-x, & 1 < x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Verify that it integrates to 1, and find $P(0.5 \leq X \leq 1.5)$.

Solution:

$$\int_0^2 f(x) \, dx = \int_0^1 x \, dx + \int_1^2 (2-x) \, dx = \frac{1}{2} + \left[2x - \frac{x^2}{2} \right]_1^2 = \frac{1}{2} + (4-2) - (2-\frac{1}{2}) = 1.$$

$$\begin{aligned}
 P(0.5 \leq X \leq 1.5) &= \int_{0.5}^1 x \, dx + \int_1^{1.5} (2-x) \, dx \\
 &= \left[\frac{x^2}{2} \right]_{0.5}^1 + \left[2x - \frac{x^2}{2} \right]_1^{1.5} = \frac{1}{2} - \frac{1}{8} + \left(3 - \frac{2.25}{2} \right) - \left(2 - \frac{1}{2} \right) = \frac{3}{8} + 1.875 - 1.5 = 0.75.
 \end{aligned}$$

Solved Example Exponential PDF

The lifetime T (in years) of a component has PDF

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

Show that it is a valid PDF, and find $P(T > 2)$ when $\lambda = 0.5$.

Solution:

$$\int_0^{\infty} \lambda e^{-\lambda t} \, dt = [-e^{-\lambda t}]_0^{\infty} = 1.$$

For $\lambda = 0.5$,

$$P(T > 2) = \int_2^{\infty} 0.5 e^{-0.5t} \, dt = [-e^{-0.5t}]_2^{\infty} = e^{-1} \approx 0.3679.$$

5 Cumulative Distribution Function (CDF)

The **cumulative distribution function** $F(x)$ of a random variable X gives the probability that X is at most x :

$$F(x) = P(X \leq x).$$

5.1 CDF for Discrete Random Variables

$$F(x) = \sum_{t \leq x} p(t).$$

Solved Example CDF of two-coin toss

For X = number of heads in two tosses, find the CDF.

Solution: We have $p(0) = \frac{1}{4}$, $p(1) = \frac{1}{2}$, $p(2) = \frac{1}{4}$.

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{1}{4}, & 0 \leq x < 1 \\ \frac{1}{4} + \frac{1}{2} = \frac{3}{4}, & 1 \leq x < 2 \\ \frac{1}{4} + \frac{1}{2} + \frac{1}{4} = 1, & x \geq 2 \end{cases}$$

Solved Example CDF from a non-uniform discrete PMF

Let X have PMF $p(0) = 0.2$, $p(1) = 0.5$, $p(2) = 0.3$. Find the CDF.

Solution:

$$F(x) = \begin{cases} 0, & x < 0, \\ 0.2, & 0 \leq x < 1, \\ 0.7, & 1 \leq x < 2, \\ 1.0, & x \geq 2. \end{cases}$$

5.2 CDF for Continuous Random Variables

$$F(x) = \int_{-\infty}^x f(t) dt.$$

Conversely, $f(x) = F'(x)$ where the derivative exists.

Solved Example CDF from PDF

For $f(x) = 2x$, $0 \leq x \leq 1$, find the CDF.

Solution:

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0, & x < 0 \\ \int_0^x 2t dt = x^2, & 0 \leq x < 1 \\ 1, & x \geq 1 \end{cases}$$

Solved Example CDF to PDF

Given the CDF

$$F(x) = \begin{cases} 0, & x < 0 \\ x^2/4, & 0 \leq x \leq 2 \\ 1, & x > 2 \end{cases}$$

find the PDF.

Solution: Differentiating:

$$f(x) = F'(x) = \begin{cases} 0, & x < 0 \\ x/2, & 0 \leq x \leq 2 \\ 0, & x > 2 \end{cases}$$

So $f(x) = x/2$, $0 \leq x \leq 2$.

Solved Example Finding CDF of piecewise PDF

Let X have PDF

$$f(x) = \begin{cases} \frac{1}{2}, & 0 \leq x \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

Find $F(x)$ and compute $P(0.5 < X < 1.5)$.

Solution:

$$F(x) = \int_{-\infty}^x f(t) dt = \begin{cases} 0, & x < 0, \\ \frac{x}{2}, & 0 \leq x \leq 2, \\ 1, & x > 2. \end{cases}$$

$$P(0.5 < X < 1.5) = F(1.5) - F(0.5) = \frac{1.5}{2} - \frac{0.5}{2} = 0.75 - 0.25 = 0.5.$$

6 Properties of CDF

- $0 \leq F(x) \leq 1$ for all x .
- $F(x)$ is non-decreasing: if $x_1 \leq x_2$, then $F(x_1) \leq F(x_2)$.
- $\lim_{x \rightarrow -\infty} F(x) = 0$ and $\lim_{x \rightarrow \infty} F(x) = 1$.
- $P(a < X \leq b) = F(b) - F(a)$.
- For continuous random variables, $P(a < X < b) = P(a \leq X \leq b) = F(b) - F(a)$.

7 Expectation (Expected Value)

7.1 Motivation

Suppose an engineer measures the resistance of 1000 resistors of the same nominal value. The individual measurements vary, but the average gives a single representative number. In probability, the **expected value** or **mean** of a random variable plays this role: it is the long-run average value of the random variable over infinitely many repetitions of the experiment.

7.2 Definition for Discrete Random Variables

If X is discrete with PMF $p(x)$, the expected value is

$$E[X] = \mu = \sum_x x p(x)$$

provided the sum converges absolutely.

Solved Example Expected value of two-coin toss

Let X = number of heads in two tosses. Find $E[X]$.

Solution:

$$E[X] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} = 0 + \frac{1}{2} + \frac{1}{2} = 1.$$

On average, we expect 1 head in two tosses.

Solved Example Expected value of die roll

Let Y = face value of a fair die. Find $E[Y]$.

Solution:

$$E[Y] = 1 \cdot \frac{1}{6} + 2 \cdot \frac{1}{6} + \cdots + 6 \cdot \frac{1}{6} = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = \frac{21}{6} = 3.5.$$

Solved Example Expected value of a custom discrete distribution

Given PMF $p(0) = 0.1, p(1) = 0.3, p(2) = 0.4, p(3) = 0.2$. Find $E[X]$.

Solution:

$$E[X] = 0(0.1) + 1(0.3) + 2(0.4) + 3(0.2) = 0 + 0.3 + 0.8 + 0.6 = 1.7.$$

Solved Example Expected value of X^2 for the above

Find $E[X^2]$ for the same PMF.

Solution:

$$E[X^2] = 0^2(0.1) + 1^2(0.3) + 2^2(0.4) + 3^2(0.2) = 0 + 0.3 + 1.6 + 1.8 = 3.7.$$

7.3 Definition for Continuous Random Variables

If X is continuous with PDF $f(x)$, the expected value is

$$E[X] = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

provided the integral converges absolutely.

Solved Example Expected value from PDF

For $f(x) = 2x, 0 \leq x \leq 1$, find $E[X]$.

Solution:

$$E[X] = \int_0^1 x \cdot 2x dx = 2 \int_0^1 x^2 dx = 2 \cdot \frac{1}{3} = \frac{2}{3} \approx 0.667.$$

Solved Example Expected lifetime

The lifetime T (in years) of a component has PDF $f(t) = e^{-t}$, $t \geq 0$. Find the expected lifetime.

Solution:

$$E[T] = \int_0^{\infty} t e^{-t} dt = \Gamma(2) = 1! = 1.$$

Average lifetime is 1 year.

Solved Example Expected value of uniform distribution

For X uniform on $[a, b]$, find $E[X]$.

Solution:

$$E[X] = \int_a^b x \cdot \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}.$$

Solved Example Expected value of triangular distribution

For the triangular PDF of the earlier example, find $E[X]$.

Solution:

$$\begin{aligned} E[X] &= \int_0^1 x \cdot x dx + \int_1^2 x(2-x) dx = \int_0^1 x^2 dx + \int_1^2 (2x - x^2) dx \\ &= \frac{1}{3} + \left[x^2 - \frac{x^3}{3} \right]_1^2 = \frac{1}{3} + \left(4 - \frac{8}{3} \right) - \left(1 - \frac{1}{3} \right) = \frac{1}{3} + \frac{4}{3} - \frac{2}{3} = 1. \end{aligned}$$

8 Expectation of Functions of Random Variables

If $g(X)$ is any function of X , then: **Discrete:**

$$E[g(X)] = \sum_x g(x)p(x).$$

Continuous:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x) dx.$$

Solved Example Expectation of X^2 (discrete)

For X = number of heads in two tosses, find $E[X^2]$.

Solution:

$$E[X^2] = 0^2 \cdot \frac{1}{4} + 1^2 \cdot \frac{1}{2} + 2^2 \cdot \frac{1}{4} = 0 + \frac{1}{2} + 1 = 1.5.$$

Solved Example Expectation of X^2 (continuous)

For $f(x) = 2x$, $0 \leq x \leq 1$, find $E[X^2]$.

Solution:

$$E[X^2] = \int_0^1 x^2 \cdot 2x \, dx = 2 \int_0^1 x^3 \, dx = 2 \cdot \frac{1}{4} = \frac{1}{2}.$$

Solved Example Expectation of a linear function

If $E[X] = 3$, find $E[2X + 5]$.

Solution:

$$E[2X + 5] = 2E[X] + 5 = 2 \cdot 3 + 5 = 11.$$

9 Properties of Expectation

- **Linearity:** For any constants a, b , $E[aX + b] = aE[X] + b$.
- $E[c] = c$ for any constant c .
- $E[X + Y] = E[X] + E[Y]$ for any two random variables X, Y .

10 Variance and Standard Deviation

The **variance** of a random variable X measures how much the values of X spread out around the mean:

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2].$$

The **standard deviation** is $\sigma = \sqrt{\text{Var}(X)}$.

10.1 Computational Formula

$$\text{Var}(X) = E[X^2] - (E[X])^2.$$

This identity is often the easiest way to compute variance.

10.2 Variance for Discrete Random Variables

$$\sigma^2 = \sum_x (x - \mu)^2 p(x).$$

10.3 Variance for Continuous Random Variables

$$\sigma^2 = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx.$$

Solved Example Variance of two-coin toss

For X = number of heads in two tosses, find the variance.

Solution: We already know $E[X] = 1$ and $E[X^2] = 1.5$.

$$\text{Var}(X) = E[X^2] - (E[X])^2 = 1.5 - 1^2 = 0.5.$$

So $\sigma = \sqrt{0.5} \approx 0.707$.

Solved Example Variance of die roll

For Y = face value of a fair die, find $\text{Var}(Y)$.

Solution: First $E[Y^2]$:

$$E[Y^2] = \frac{1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2}{6} = \frac{91}{6} \approx 15.1667.$$

Then:

$$\text{Var}(Y) = E[Y^2] - (E[Y])^2 = \frac{91}{6} - \left(\frac{21}{6}\right)^2 = \frac{91}{6} - \frac{441}{36} = \frac{546 - 441}{36} = \frac{105}{36} = \frac{35}{12} \approx 2.917.$$

Solved Example Variance from PDF

For $f(x) = 2x$, $0 \leq x \leq 1$, find the variance.

Solution: We have $E[X] = 2/3$.

$$E[X^2] = \int_0^1 x^2 \cdot 2x dx = 2 \int_0^1 x^3 dx = 2 \cdot \frac{1}{4} = \frac{1}{2}.$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{9 - 8}{18} = \frac{1}{18} \approx 0.0556.$$

Solved Example Variance of uniform distribution

For X uniform on $[a, b]$, find $\text{Var}(X)$.

Solution: We know $E[X] = \frac{a+b}{2}$. Need $E[X^2]$:

$$E[X^2] = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{b^2 + ab + a^2}{3}.$$

Then

$$\text{Var}(X) = \frac{b^2 + ab + a^2}{3} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}.$$

Solved Example Variance of exponential distribution

For $f(t) = \lambda e^{-\lambda t}$, $t \geq 0$, find $\text{Var}(T)$.

Solution: We know $E[T] = 1/\lambda$. Need $E[T^2]$:

$$E[T^2] = \int_0^\infty t^2 \lambda e^{-\lambda t} dt = \frac{2}{\lambda^2}.$$

Then

$$\text{Var}(T) = \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{1}{\lambda^2}.$$

11 Properties of Variance

- $\text{Var}(X) \geq 0$ always.
- $\text{Var}(X) = 0$ if and only if X is a constant.
- $\text{Var}(aX + b) = a^2 \text{Var}(X)$.
- For independent random variables X and Y , $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

Solved Example Variance under linear transformation

If $\text{Var}(X) = 4$, find $\text{Var}(3X + 2)$.

Solution:

$$\text{Var}(3X + 2) = 3^2 \text{Var}(X) = 9 \times 4 = 36.$$

Note: adding a constant 2 does not change the variance.

12 Relation Between Expectation and Variance

The key identity that connects expectation and variance is:

$$\boxed{\text{Var}(X) = E[X^2] - (E[X])^2}.$$

This can be derived as follows:

$$\text{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2.$$

Since $\mu = E[X]$,

$$= E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2 = E[X^2] - (E[X])^2.$$

This relationship is used repeatedly in problems.

13 Summary of Key Formulas

- **PMF (discrete):** $p(x) = P(X = x)$, $\sum p(x) = 1$.
- **PDF (continuous):** $P(a \leq X \leq b) = \int_a^b f(x)dx$, $\int f = 1$.
- **CDF:** $F(x) = P(X \leq x)$. Discrete: $\sum_{t \leq x} p(t)$; Continuous: $\int_{-\infty}^x f(t)dt$.
- **Expectation:** Discrete: $\sum xp(x)$; Continuous: $\int xf(x)dx$.
- **Variance:** $\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$.

14 Practice Problems

1. A coin is biased so that $P(H) = 0.7$. Let X be the number of heads in 3 tosses. Find the PMF of X and sketch it.
2. Verify whether $p(x) = k(1/2)^x$, $x = 1, 2, 3, \dots$, can be a PMF; if so, find k .
3. A random variable has $f(x) = c(1 - x^2)$, $-1 \leq x \leq 1$. Find c , $P(0 \leq X \leq 0.5)$, and the CDF.
4. The CDF of a random variable is $F(x) = 1 - e^{-2x}$, $x \geq 0$. Find the PDF and compute $P(0.5 \leq X \leq 1)$.
5. A die is rolled. Let X = face value. Compute $E[X]$, $E[X^2]$, and $\text{Var}(X)$.
6. A random variable has PDF $f(x) = 3x^2$, $0 \leq x \leq 1$. Find $E[X]$, $E[X^2]$, and $\text{Var}(X)$.
7. If $E[X] = 5$ and $\text{Var}(X) = 4$, find $E[2X - 3]$ and $\text{Var}(2X - 3)$.
8. A lot of 10 items contains 3 defectives. Three items are drawn without replacement. Let X be the number of defectives. Find the PMF of X .